Week 2 Deliverable: EDA + Preprocessing

Team: MachineLearners

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Objective

The goal of this notebook is to explore the Telco Customer Churn dataset, identify patterns related to customer behavior, handle missing values, and prepare the data for machine learning models.

Prediction Task

Target variable: `Churn` This is a **classification problem**, where we predict whether a customer will leave the company.

Hypothesis

We hypothesize that customers with month-to-month contracts and higher monthly charges are more likely to churn compared to customers with long-term contracts and lower charges.

Imports

In [1]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
```

Load dataset

The dataset contains 7043 customers and multiple features including demographic, service, and billing information. The target variable is `Churn`, which indicates whether the customer left the company. Some columns such as `TotalCharges` contain non-numeric values and require cleaning before analysis.

In [2]:
```
df = pd.read_csv("Telco-Customer-Churn.csv")

# FIX DATA
df["TotalCharges"] = pd.to_numeric(df["TotalCharges"], errors="coerce")
```

```
df.head()
```

Out[2]:

	customerID	gender	SeniorCitizen	Partner	Dependents	tenure	PhoneService	MultipleLines	Inte
0	7590-VHVEG	Female	0	Yes	No	1	No	No phone service	
1	5575-GNVDE	Male	0	No	No	34	Yes	No	
2	3668-QPYBK	Male	0	No	No	2	Yes	No	
3	7795-CFOCW	Male	0	No	No	45	No	No phone service	
4	9237-HQITU	Female	0	No	No	2	Yes	No	

5 rows × 21 columns

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1. Dataset Overview

We begin by examining the dataset structure, data types, missing values, and summary statistics.

In [3]:

```
print("Shape:", df.shape)
print("\nData types:")
print(df.dtypes)

print("\nMissing values:")
print(df.isnull().sum())

display(df.describe())
```

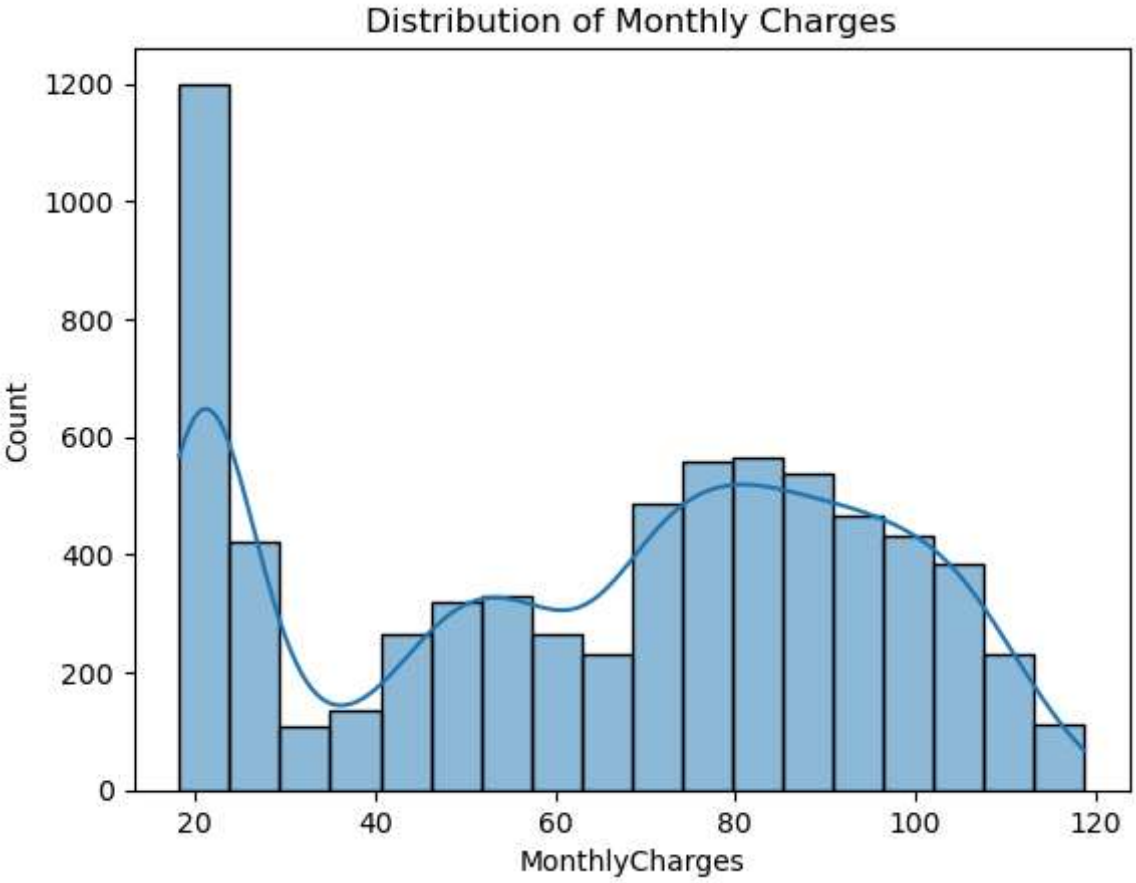
```
Shape: (7043, 21)

Data types:
customerID           object
gender               object
SeniorCitizen         int64
Partner              object
Dependents           object
tenure                int64
PhoneService         object
MultipleLines        object
InternetService      object
OnlineSecurity       object
OnlineBackup         object
DeviceProtection     object
TechSupport          object
StreamingTV          object
StreamingMovies      object
Contract             object
PaperlessBilling     object
PaymentMethod        object
MonthlyCharges      float64
TotalCharges        float64
Churn                object
dtype: object

Missing values:
customerID           0
gender               0
SeniorCitizen        0
Partner              0
Dependents           0
tenure               0
PhoneService         0
MultipleLines        0
InternetService      0
OnlineSecurity       0
OnlineBackup         0
DeviceProtection     0
TechSupport          0
StreamingTV          0
StreamingMovies      0
Contract             0
PaperlessBilling     0
PaymentMethod        0
MonthlyCharges       0
TotalCharges        11
Churn                0
dtype: int64
```

	SeniorCitizen	tenure	MonthlyCharges	TotalCharges
count	7043.000000	7043.000000	7043.000000	7032.000000
mean	0.162147	32.371149	64.761692	2283.300441
std	0.368612	24.559481	30.090047	2266.771362
min	0.000000	0.000000	18.250000	18.800000
25%	0.000000	9.000000	35.500000	401.450000
50%	0.000000	29.000000	70.350000	1397.475000
75%	0.000000	55.000000	89.850000	3794.737500
max	1.000000	72.000000	118.750000	8684.800000

2. Visualisations (6)

1) Histogram (MonthlyCharges)

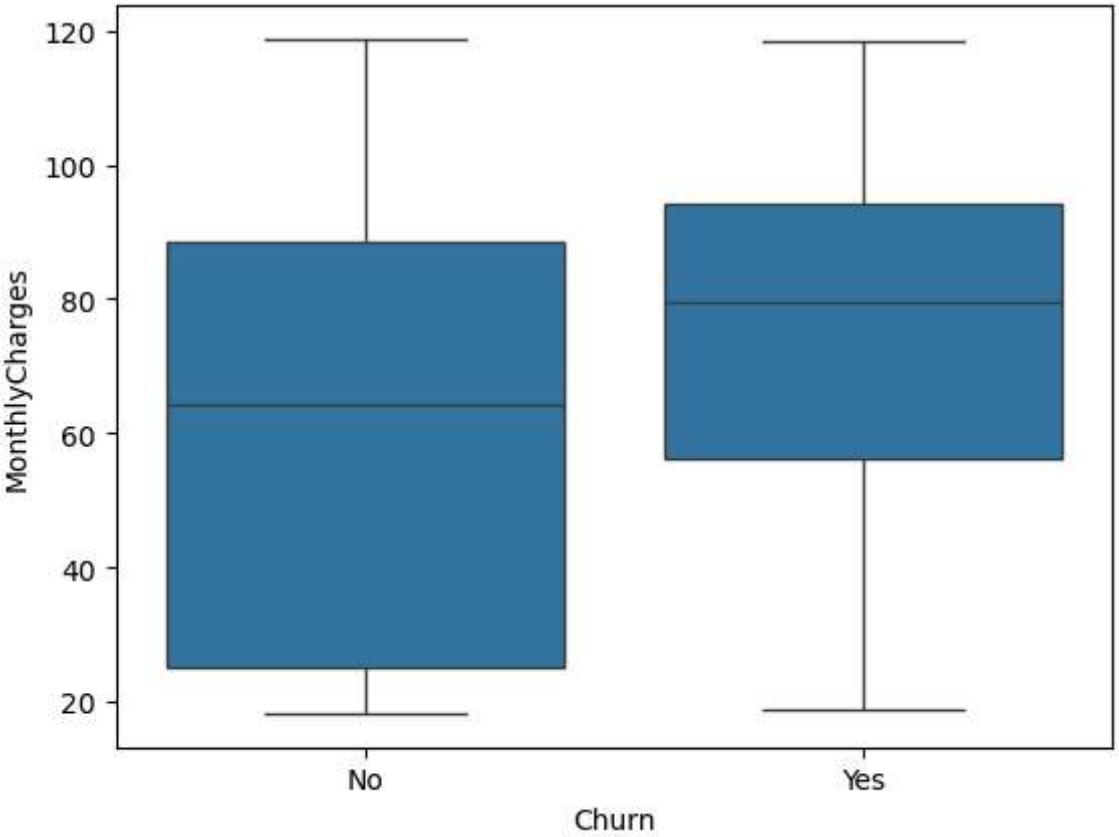
In [4]:
```
sns.histplot(df["MonthlyCharges"], kde=True)
plt.title("Distribution of Monthly Charges")
plt.show()
```



Monthly charges are spread across a wide range, with a slight concentration in mid to high values. Higher charges may be associated with increased churn, supporting our hypothesis.

2) Boxplot (MonthlyCharges vs Churn)

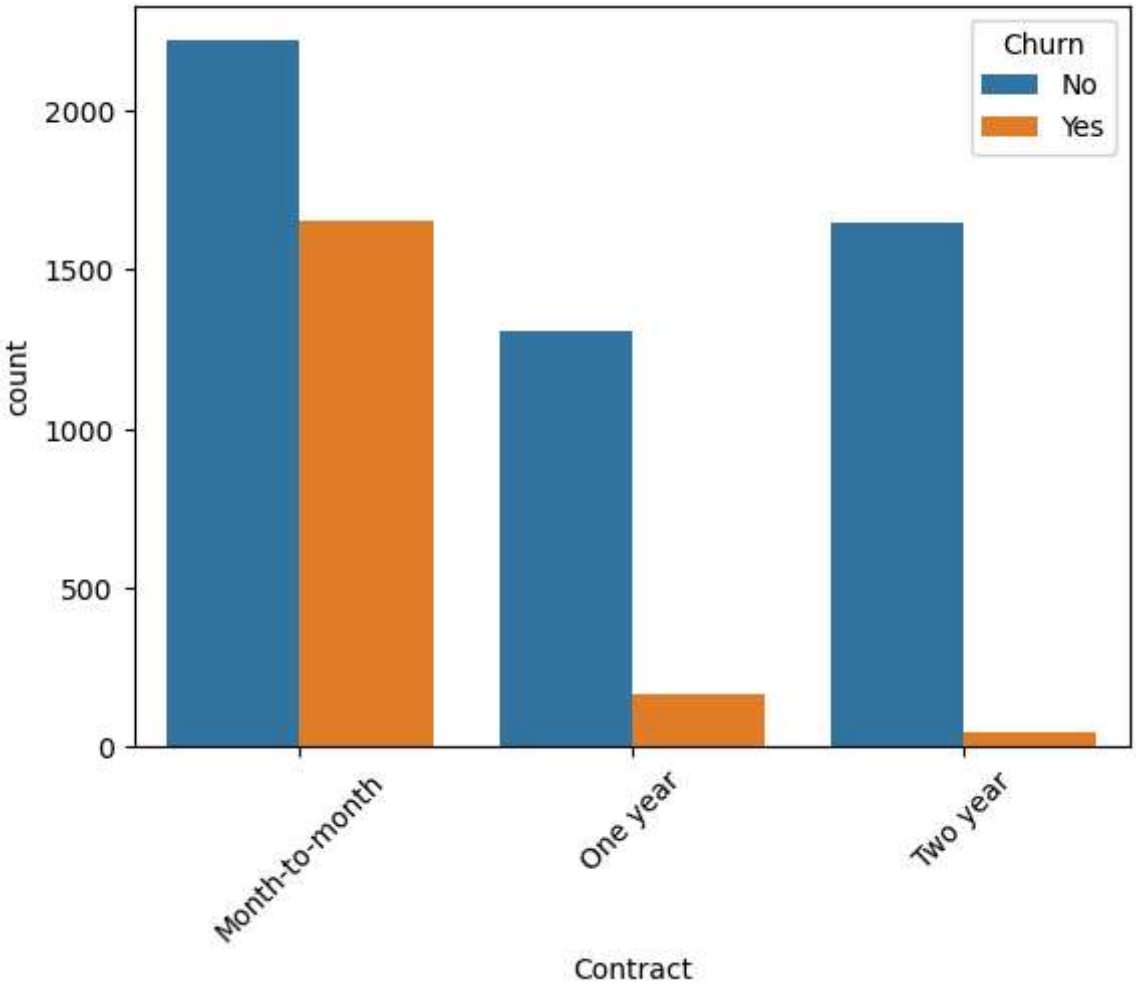
In [5]:
```
sns.boxplot(data=df, x="Churn", y="MonthlyCharges")
plt.show()
```



Customers who churn tend to have higher monthly charges on average. This supports our hypothesis that higher cost may lead to dissatisfaction and churn.

3) Countplot (Contract vs Churn)

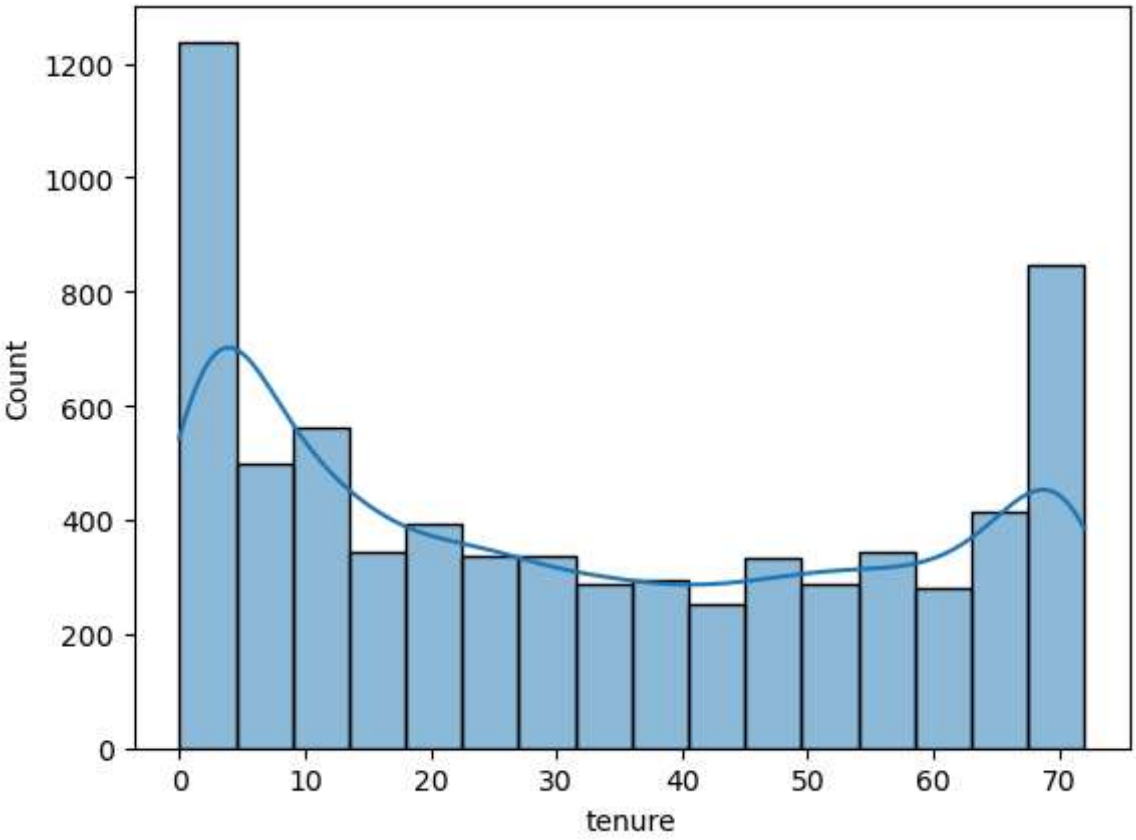
In [6]:
```
sns.countplot(data=df, x="Contract", hue="Churn")
plt.xticks(rotation=45)
plt.show()
```



Customers with month-to-month contracts show significantly higher churn rates compared to those with longer-term contracts, strongly supporting our hypothesis.

4. Histogram (tenure)

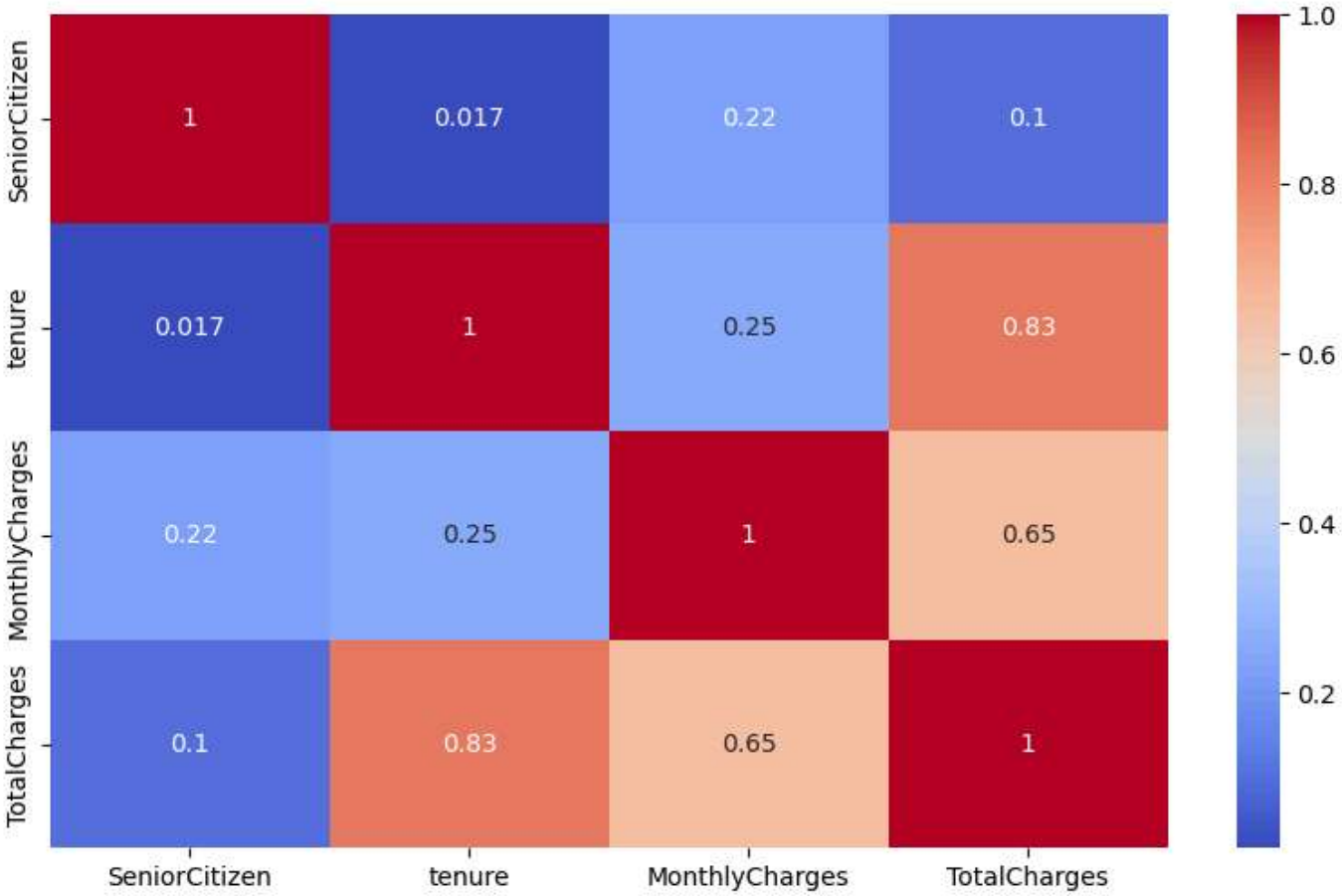
In [7]:
```
sns.histplot(df["tenure"], kde=True)
plt.show()
```



Customers with short tenure are more common among churned users. This suggests that new customers are more likely to leave early.

5. Correlation heatmap

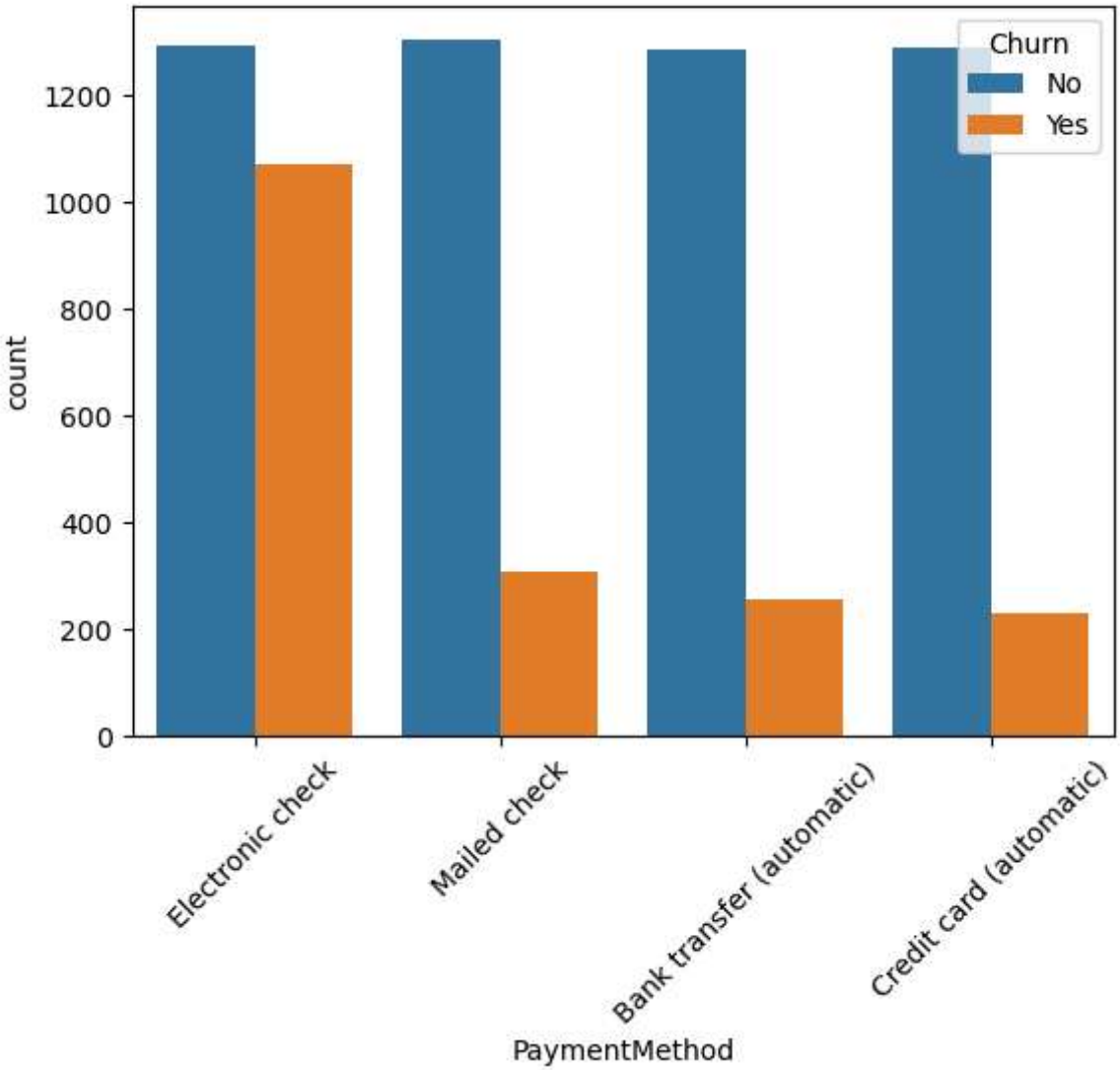
In [8]:
```
plt.figure(figsize=(10,6))
corr = df.select_dtypes(include=np.number).corr()
sns.heatmap(corr, annot=True, cmap="coolwarm")
plt.show()
```



The strongest correlations are between `tenure` and `TotalCharges`, and between `MonthlyCharges` and `TotalCharges`, which is expected because total charges accumulate over time and depend on the monthly bill. In relation to churn, customers with shorter tenure and higher monthly charges appear more likely to leave, which supports our hypothesis. These variables are therefore likely to be important predictors in the classification model.

6. PaymentMethod vs Churn

In [9]:
```
sns.countplot(data=df, x="PaymentMethod", hue="Churn")
plt.xticks(rotation=45)
plt.show()
```



Certain payment methods show higher churn rates, which may indicate differences in customer reliability or satisfaction levels.

---

3. Missing Value Handling

We identify missing values and handle them using column-specific strategies instead of a blanket drop.

In [10]:
```
df.isnull().sum()
```

Out[10]:
```
customerID           0
gender               0
SeniorCitizen        0
Partner              0
Dependents           0
tenure               0
PhoneService         0
MultipleLines        0
InternetService      0
OnlineSecurity       0
OnlineBackup         0
DeviceProtection     0
TechSupport          0
StreamingTV          0
StreamingMovies      0
Contract             0
PaperlessBilling     0
PaymentMethod        0
MonthlyCharges       0
TotalCharges        11
Churn                0
dtype: int64
```

In [11]:
```
# Only TotalCharges has missing values
df["TotalCharges"] = df["TotalCharges"].fillna(df["TotalCharges"].median())
```

The `TotalCharges` column contained missing values due to incorrect data type conversion. Since it is an important numerical feature, we imputed missing values using the median, which is robust to outliers.

Drop useless column

In [12]:
```
df.drop("customerID", axis=1, inplace=True)
```

The `customerID` column is an identifier and does not contribute to prediction, so it was removed.

Encoding

In [13]:
```
df["Churn"] = df["Churn"].map({"Yes":1, "No":0})
```

In [14]:
```
df = pd.get_dummies(df, drop_first=True)
```

We used one-hot encoding for categorical variables because most of them are nominal and have relatively low cardinality. Since our planned models include standard machine learning classifiers that work with numerical input, one-hot encoding is appropriate because it avoids creating false ordinal relationships between categories.

The categorical variables have relatively low cardinality, making one-hot encoding an efficient and suitable approach.

Feature engineering

In [15]:
```
df["AvgCharge"] = df["TotalCharges"] / (df["tenure"] + 1)
```

We created a new feature `AvgCharge`, which represents average spending per month. This may better capture customer behavior than total charges alone.

In [16]:
```
df.to_csv("clean_telco_data.csv", index=False)
```

Conclusion

In this notebook, we explored the Telco Customer Churn dataset and identified key factors influencing customer churn. We visualized important relationships, handled missing values using appropriate strategies, encoded categorical variables, and engineered new features. Finally, we prepared the dataset for modelling by splitting it into training and test sets and applying scaling. This ensures a clean and reliable input for machine learning models.

Split

In [24]:
```
X = df.drop("Churn", axis=1)
y = df["Churn"]
```

In [23]:
```
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

#Scaling

scaler = StandardScaler()
num_cols = X_train.select_dtypes(include=np.number).columns

#We used StandardScaler to normalize numerical features. Importantly, the scaler was fitted o



X_train[num_cols] = scaler.fit_transform(X_train[num_cols])
X_test[num_cols] = scaler.transform(X_test[num_cols])



X_train.to_csv("X_train.csv", index=False)
X_test.to_csv("X_test.csv", index=False)
y_train.to_csv("y_train.csv", index=False)
y_test.to_csv("y_test.csv", index=False)

print("X_train shape:", X_train.shape)
print("X_test shape:", X_test.shape)
print("y_train shape:", y_train.shape)
print("y_test shape:", y_test.shape)
```

```
X_train shape: (5634, 31)
X_test shape: (1409, 31)
y_train shape: (5634,)
y_test shape: (1409,)
```